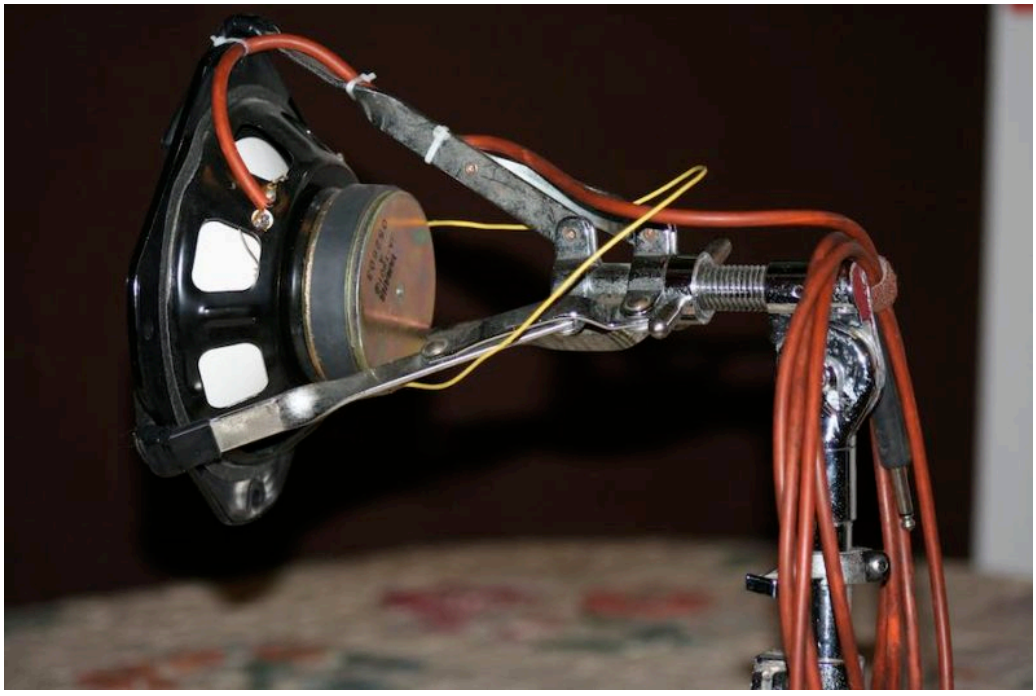
in **Gear , Microphones** by Bobby Owsinski

DIY Subkick Info – Bobby Owsinski's Music Production Blog



The subkick microphone phenomenon has been with us for a while now and it seems like you won't find a tracking session without one anymore. It all started due to the burning desire to get more bottom end from the kick drum without having to crank up the low-end EQ. That being said, the trend is not a new idea by any means, as engineer **Geoff Emerick** first tried this on **Beatles** records ("Rain" and "Paperback Writer") in the '60s using a speaker cabinet instead of the raw speaker that we see today.

While you can certainly buy a pre-made subkick microphone (**Yamaha** makes the **SKRM-100** – which is actually a speaker mounted inside a drum shell, the **DW Moon Mic, or Solomon Mics LofReQ**), you can build one a lot cheaper. Here's how to do it, but be aware that we'll get to some caveats a bit later.

1. **Find a speaker that you want to use.** We'll at the type in a second. Power doesn't matter, nor does the size (although you might want to experiment a bit here).

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2. Add a cable, or even speaker wire. It doesn't much matter since it won't be too long to affect the sound and the level is going to be really strong anyway.

3. Wire pin 2 of a male XLR connector to the positive terminal of the speaker and pin 3 of the connector to the negative terminal. If you're using a microphone cable you can either attach the ground to the basket of the speaker or just cut it off and not use it at all. Some people wire the cable to a male 1/4" guitar plug, then connect that to a direct box (see #5).

4. Mount the speaker. This may be the easiest or most difficult thing to do. Some people just let the magnet of the speaker attach itself to a mic stand, and if it's strong enough, it will just stay where it is during tracking. A better way is to use a snare stand to to cradle it (see the graphic on the left).

5. Make sure to add a pad. The level is going to be hot, so you'll need a to insert either a 20dB inline pad, the pad on the preamp, or connect it to the line input of the preamp to keep it from overloading. One of the reasons why some use the direct box method is for the built-in pad if it has one.

Okay, now for the caveats. The speaker is really a big deal, but not for the reason why you might think. **Hugh Robjohns** covered this nicely in a **Sound On Sound** article a while back, but the idea is that the size of the speaker doesn't matter nearly as much as its free-air resonance. That's what's actually generating a good deal of the subkick sound you hear, and it's also why some speakers sound way better than others.

The original subkicks were made from the woofer from a Yamaha **NS-10M** monitor because there were lots of replacement drivers lying around at the time. It turns out that an NS-10 woofer happens to have a free-air resonance at the right frequency and the right damping so that it resonates nicely in front of a kick drum. The problem is that Yamaha no longer produces the NS-10, and the factory that made the woofer has closed.

Yamaha tried to capitalize on the subkick craze by manufacturing their SKRM 100 model that contains a 6-1/2-inch speaker mounted inside a 7-ply maple shell with black mesh heads, so it's actually a speaker mounted inside a drum shell. The problem is that it never sounded the same as the homemade version with the NS-10, and has since been discontinued.

Since then a number of new subkick mics have been produced to fill the void. Drum manufacturer DW sells The Moon subkick mic, Avantone has its Pro Kick, Solomon Mics has its LoFreq. Each manufacturer approaches the design a little differently, but they're all pretty much the same as the DIY mic that you can do yourself.

The major advantage of buying one of these pre-built mics is that you won't have to worry about the subkick wiring or coming up with a way to mount it

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on a stand, which may be the most difficult part of the entire DIY subkick operation. All subkick mics that you can buy come with their own stand.

No matter, if you don't have a subkick and want to add one on the cheap, just find a old woofer that's lying around (it doesn't have to be larger than 8" because bigger is not necessarily better), follow the directions above, and you'll soon have more low end than you know what do with.

Using The Subkick

- To use the subkick, get the sound of your main kick drum first and then add about 10 percent of the subkick to the main kick, or just until you hear it. Of course, if your speakers won't reproduce anything as low as 30 or 40Hz, then you probably shouldn't even try this because you'll just be guessing about how loud the subkick should actually be, and if you add too much the kick will sound too woofy and will lack definition.

- You can use the subkick on other instruments as well. Try it on a bass amp or even a trombone.

A subkick can be the best addition you ever made to your drum recording setup or it can create a mad jumble of low-end that's out of control. The secret is that you don't need much of it. Use it wisely!

People also ask:

How do you make a Subkick?
Find a raw speaker driver that you're not using, attach a short cable to the speaker connected to a female XLR or ¼ inch jack, and mount it on a stand (a snare stand works well).

What does a Subkick Do?
It captures the extreme low frequencies that most microphones don't pick up. You can also record it to a separate track to have extra control over the low end.

Do I really need to use a Subkick?
No, engineers and producers have been recording excellent kick drum sounds for decades without it. But if you want or need lower frequencies than you're capturing, this is an excellent way to do it.

How do you turn a speaker into a microphone?
A speaker and a microphone both work on the same principle. They're dynamic transducers with the only differences being that they're designed for a particular application. For a speaker to work as a microphone, you just have to wire it like a mic and feed it into a microphone preamp like you would with any other microphone.

Does a Subkick need phantom power?

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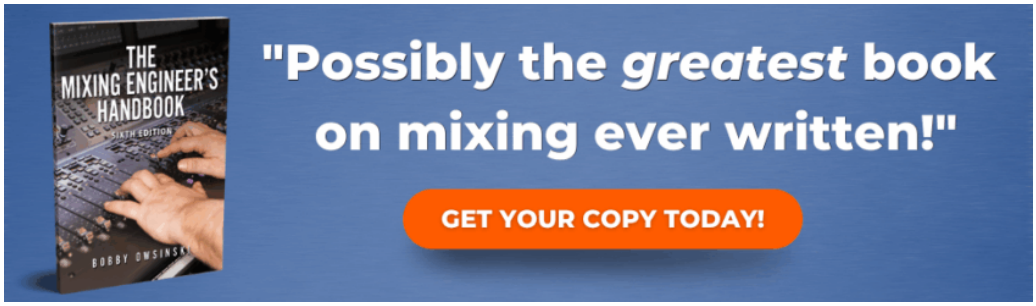
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No, a subkick is just a passive transducer and therefore doesn't need an outside power source for it to work. Think of it like a big Shure SM57, which doesn't need phantom power either.

You can read more from ***The Recording Engineer's Handbook*** and my other books on the excerpt section of bobbyowsinski.com. Also, everything you wanted to know about alternative mixes.



#DW #SUBKICK #THE BEATLES #YAMAHA

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